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AIRs - LM in

Business Mathematics

Week 1-2: Module 1

GOVERNMENT PROPERTY
NOT FOR SALE

Business Mathematics

Week 1-2: Module 1

Target

These days, in most business computations, converting fraction to decimal, fraction to percent, decimal to fraction, or percent to fraction are some of the basic practical knowledge in solving business problems. The knowledge that you will gain from this module will prepare you better in your study of the next lesson.

This module will provide you with information and activities that will help you understand Fractions, Decimals, and Percent.

After going through this module, you are expected to:

1. Express:
   a. fractions to decimals and percent forms
   b. decimals to fractions and percent forms
   c. percent to fractions and decimal forms
2. Give real life situations to illustrate fractions, decimals, and percent
3. Solve Problems involving fractions, decimals and percent

Before going on, check how much you know about this topic. Answer the pretest on the next page in a separate sheet of paper.

Module 1 Converting Fractions, Decimals and Percent

Jumpstart

Activity 1. Fill the Missing Part!

Directions: Fill in the missing fraction, decimal, or percent. Write fractions in simplest form and round off numbers up to two decimal point. Show a neat and complete solution on the space provided.

No.	FRACTION	DECIMAL	PERCENT
1	$\frac{7}{23}$		
2		0.346	
3			53%
4	$\frac{12}{15}$		
5		0.26	
6			32.5%
7	$\frac{3}{19}$		
8		1.25	
9			$85\frac{1}{2}\%$
10	$\frac{2}{9}$		

YOUR SOLUTION STARTS HERE...

Discover

The following are the rules in converting fractions to decimals to percent and vice versa:

A. Converting Fraction to Decimal and Percent Forms

Step 1: Change the fraction to decimal by simply performing division (Fraction to Decimal)
Step 2: Move the decimal point two places to the right
Step 3: Affix the percent sign (Decimal to Percent)

B. Converting Decimal to Fraction Forms

Step 1: Move the decimal point two places to the right
Step 2: Place a denominator of the resulting number in multiples of 10 (that is 10, 100, 1000 etc.)
Step 3: Reduce the fraction to lowest term

C. Converting Percent to Decimal

Step 1: Remove the percent sign
Step 2: Move the decimal point two places to the left

D. Converting Percent to Fraction

Step 1: Remove the percent sign
Step 2: Place a denominator of the resulting number in multiples of 10 (that is 10, 100, 1000 etc.)
Step 3: Reduce the fraction to lowest term (Decimal to Fraction)

The following are some examples in converting fractions to decimal, fractions to percent, decimal to fraction, or percent to fraction.

A. Converting Fraction to Decimal and Percent Forms

Examples:

1. $\frac{2}{5}$ STEPS:

   → 0.4 (change the fraction to decimal by simply performing division)

$\rightarrow 40$ (move the decimal point two places to the right)

$= 40\%$ (affix the percent sign)

2. $\frac{5}{16} = 0.3125 = 31.25\%$
3. $\frac{3}{10} = 0.3 = 30.\%$

B. Converting Decimal to Fraction Forms

Examples:

1. 0.65 STEPS:

$\rightarrow 65$ (move the decimal point two places to the right)

$= \frac{65}{100}$ (place a denominator of the resulting number in multiple of 100)

$= \frac{65 \div 5}{100 \div 5}$ (reduce the fraction to lowest term)

$= \frac{13}{20}$ (answer)

2. $0.175 = \frac{175 \div 25}{1000 \div 25} = \frac{7}{20}$
3. $0.0242 = \frac{242 \div 2}{10000 \div 2} = \frac{121}{5000}$

C. Converting Percent to Decimal

Examples:

1. 23% STEPS:

$\rightarrow 23$ (remove the percent sign)

$= 0.23$ (move the decimal point two places to the left)

2. $57.2\% = 0.572$
3. $0.05\% = 0.0005$

D. Converting Percent to Fraction

Examples:

1. 25% STEPS:

$\rightarrow 25$ (remove the percent sign)

$= \frac{25}{100}$ (place a denominator of the resulting number in multiple of 100)

$= \frac{25 \div 5}{100 \div 5}$ (reduce the fraction to lowest term)

$= \frac{1}{4}$ (answer)

2. $53\% = \frac{53}{100}$
3. $12.5\% = \frac{125 \div 125}{1000 \div 125} = \frac{1}{8}$

5 (Very Good)	4(Good)	3 (Fair)	2 (Poor)	1 (Needs Improvement)
The answer shows a deep understanding of mathematical concepts as seen by the numerous strategies and concepts used to solve the problem. The justifications are correct with correct, complete and organized computations.	The answer shows a deep understanding of mathematical concepts as seen by the several strategies and concepts used to solve the problem. The justifications are correct and all the computations are correct.	The answer shows understanding of mathematical concepts as seen by few strategies and concepts used to solve the problem. The justifications are correct and almost all the computations are correct.	The answer shows a little understanding of mathematical concepts as seen by only one strategy and few concepts used to solve the problem. Most of the justifications are correct and most of the computations are correct.	There is an attempt to show understanding but the justifications are incorrect and most of the computations are incorrect.

Gauge

Activity 8. Convert Me!

Directions: Provide what is being asked. Show a neat and complete solution on a separate sheet of paper.

A. Express the following fractions to decimal. If not whole number, compute up to two decimal places and round to the nearest hundredth.

1. $\frac{3}{9}$
2. $\frac{11}{15}$
3. $\frac{17}{8}$
4. $\frac{234}{1111}$
5. $\frac{13}{11}$

B. Express the following decimals to fraction, renaming in simplest terms.

1. 0.56
2. 0.785
3. 0.4
4. 3.45
5. 12.25

C. Express the following fractions to percent.

1. $\frac{1}{7}$
2. $\frac{17}{19}$
3. $\frac{29}{31}$
4. $\frac{125}{111}$
5. $9\frac{2}{3}$

D. Express the following percent to fraction.

1. 65%
2. 78%
3. 99.9%
4. $25\frac{1}{5}\%$
5. 0.039%

E. Express the following decimals to percent.

1. 0.78
2. 0.5
3. 0.325
4. 1.95
5. 0.0052

F. Express the following percent to decimal.

1. 95%
2. 8.76%
3. 250
4. $35\frac{1}{4}\%$
5. 0.759%

Activity 9. Give A Real-Life Situation!

Direction: The first row of the following table is an example of a real-life category and its corresponding situation in representing real-life applications of fraction, decimal and percent. Give a situation for each category below to complete the table.

Category	**Real-life situation**
Example: FAST FOOD	We ate at a fast food restaurant with my friends and when the check of our orders is released, we divided up the bill using **fractions.** Since we are a group of five friends, each one of us paid $\frac{1}{5}$ of the total bill.
1. GROCERY SHOPPING	
2. SPORTS	

Business Mathematics

Week 3: Module 2

Target

Proportion is a mathematical comparison between two numbers. Often, these numbers can represent a comparison between things or people.

This module was designed to help you understand what proportion is. It contains explicit lessons on the different kinds of proportion, and set of activities that will hone your skills in applying the concept of proportion in making life decisions.

After going through this learning material, you are expected to:

1. identify the different kinds of proportions and write examples of real-life situations for each

Before going on, check how much you know about this topic. Answer the activities on the next page.

Discover

Congratulations! You've earned a badge! Time to unlock new concept!

Two ratios $\left(\frac{a}{b}, \frac{c}{d}\right)$ can be equal. Or that is $\frac{a}{b} = \frac{c}{d}$. A statement that two ratios are equal is called **proportion**. Therefore, the statement $\frac{a}{b} = \frac{c}{d}$ is a **proportion**.

Each of the four numbers in a proportion is called a **term** of the proportion.

Odds Notation	second term first term — $a:b = c:d$ — fourth term third term
Fractional Notation	first term third term $\frac{a}{b} = \frac{c}{d}$ second term fourth term

The first and fourth terms, a and d are the extreme terms or ***extremes***, while the second and third terms b and c are the middle terms or ***means.***

Example 1: In the proportion $\frac{4}{5} = \frac{12}{15}$, name the four terms, the means, and the extremes.

Solution:

First term = 4	Third term = 12	Means = 5 and 12
Second term = 5	Fourth term = 15	Extremes = 4 and 15

In order to check whether two ratios are equal or form a proportion, we apply the cross product rule. That is, the product of the means must be equal to the product of the extremes. Or, $ad = bc$.

Example 2: Let's take the given in Example 1 and test whether it is proportional or not. So, we have $\frac{4}{5} = \frac{12}{15}$. Applying cross product rule, we have $4(15) = 5(12)$. Or that is, $60 = 60$. Therefore, $\frac{4}{5} = \frac{12}{15}$ is a proportion.

Kinds of Proportion

1. **Direct Proportion**

   Two variables (x and y) vary such that as x increases, y also increases, or as x decreases, y also decreases proportionally. Or that is, the ratio $\frac{x}{y}$ is always the same. The same holds true with the ratio $\frac{y}{x}$.

   Example: Two pots of cactus cost Php 130.00. How much do 5 pots of cactus cost? What type of proportion is this?

   In this situation, we have two variables x (number of pots) and y (cost of pot). By analyzing, we can say that the more pots, the higher the cost. That is, both quantities are increasing. Therefore, this is an example of direct proportion.

2. **Indirect/Inverse Proportion**

   Two variables (x and y) vary such that as x increases, y decreases, or as x decreases, y increases proportionally. Or that is, the product of x and y is always the same.

   Example: Two men can finish planting rice in a hectare of rice field for four days. At the same rate, many days will it take for 8 men to do the job? What type of proportion is this?

   In this problem, we have two variables x (number of workers/men) and y (days to finish planting rice). And we can infer that the greater number of workers, the lesser number of days to finish the job. That is, as one quantity increases, the other quantity decreases. Therefore, this is an example of indirect/inverse proportion.

3. **Partitive Proportion**

   In this type of proportion, a whole is divided into parts.

   Example: A cash incentive worth Php 30,000.00 is to be divided by 3 families with the ratio $1:3:4$, how much does each family receives? What type of proportion is this?

   This situation shows division of whole into parts or portions. Therefore, this is an example of partitive proportion.

Explore

You're now ready to level up! Here's another set of fun activities to practice your skills!

Activity 1: Proportion or Not?

Direction: Using the cross product rule, tell whether the following are proportion or not. Show your solution.

	Proportion/Not	**Proof/Solution**
1. $\frac{18}{45} = \frac{22}{55}$		
2. $\frac{25}{60} = \frac{30}{45}$		
3. $\frac{48}{56} = \frac{60}{70}$		
4. $\frac{49}{84} = \frac{63}{108}$		
5. $\frac{64}{121} = \frac{72}{144}$		

Activity 2: Identify me!

Direction: Identify what type of proportion are the following:

_____1. mobile data used and number of days using Facebook
_____2. a roll cake sliced into 10 portions
_____3. electricity consumption and number of days using electrical appliances
_____4. number of workers and days to finish a job
_____5. a liter of lemon juice to be divided among 3 children
_____6. a pack of loaf bread to be shared by four
_____7. wage and number of working days
_____8. size of pizza and its price
_____9. number of error and score in exam
_____10. a 1.5L soda to be divided among 5 persons

Activity 3: Give me!

Direction: Give two (2) examples of real-life situation in each kind of proportion.

1. Direct Proportion (e.g. number of days and the length of hair)
   a. _____
   _____
   b. _____
   _____

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BUSINESS MATH QUARTER 3

Most Essential Learning Competencies	Output To Be Submitted	Date of Completion
Quarter III Week 1 ***2 Fractions/ decimals and percentage*** **February 21-25, 2022** ✓ Give real-life situations to illustrate fractions, decimals, and percent (ABM_BM11FO-Ic-4) ✓ Solve problems involving fractions, decimals and percent (ABM_BM11FO-Id-5)	**Worksheet for Week 1 -2 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 1-2 . Gauge. Activity 8. Convert Me! (A, B, C, D, E, F) o Write all your answers on yellow paper. o Write the activity title o Write the quarter number and week number on the top right side of the paper.	**March 3, 2022**
Quarter III Week 3 – ***Key Concepts of Ratio and Proportion*** **February 28 – March 4, 2022** ✓ Identify the different kinds of proportions and write examples of real-life situations for each	**Worksheet for Week 3 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 3. Explore. Activity 1. #s 1-5	**March 7, 2022**
Quarter III Week 4 ***– Key Concepts of Ratio and Proportion*** **March 7-11, 2022** ✓ Solve problems involving direct, inverse and partitive proportion (ABM_BM11RP-If-4)	**Worksheet for Week 4 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 4. Gauge #s 1-15	**March 14, 2022**
Quarter III Week 5 ***Key Concepts of Buying and Selling*** **March 14-18, 2022** ✓ Differentiate Mark-on, Mark-down, and Mark-up obtain Mark-on, Mark-down and Mark-up of a product (ABM_BM11BS-Ig-1)	**Worksheet for Week 5 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 5a. Explore. Activity 1. Can You Tell? #s 1-10 Module 5b, Explore. Activity 1. Complete Me! Letters – A, B, C	**March 21, 2022**
Quarter III Week 6 ***Key Concepts of Buying and Selling*** **March 21-25, 2022** ✓ Differentiate Mark-up from margins (ABM_BM11BS-Ih-3) ✓ Describe how gross margins is used in sales (ABM_BM11BS-Ih-4) ✓ Compute single trade discounts and discount series	**Worksheet for Week 6 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities Module 6. Gauge #s 1-10 Module 7. Gauge #s 1-10 Module 8. Enrichment. Activity 1. Solve Me #s 1-3	**March 28, 2022**

Quarter III Week *7 Key Concepts of Buying and Selling* **March 28- April 1, 2022** ✓ Differentiate profit from loss (ABM_BM11BS-Ii-6) ✓ Illustrate how profit is obtained and how to avoid loss in a given transaction (ABM_BM11BS-Ii-7)	**Worksheet for Week 7 (Reference Activity Sheet Quarter III)** Instructions: Answer the following activities Module 9. Explore #s 1-5 Module 10. Gauge #S 1-10	**April 4, 2022**
Quarter III Week *8 Key Concepts of Buying and Selling* **April 4-8, 2022** ✓ Determine the break-even point ✓ Solve problems involving buying and selling products (ABM_BM11BS-Ij-9)	**Worksheet for Week 8 (Reference: Activity Sheet Quarter 1II)** Instructions: Answer the following activities Module 11. Explore. Activity 1. Is it a Fact or a Bluff?	**April 11, 2022**
Quarter III Week 9 Key Concepts of Buying and Selling **April 11-13, 2022** ✓ Compute interest specifically as applied to mortgage, amortization and on services/ utilities and on deposits and loans	**Worksheet for Week 8 (Reference: Activity Sheet Quarter 1II)** Instructions: Answer the following activities Module 13. Gauge # 1-15	**April 18, 2022**